

Glossary

Simple definitions for classification, Decision Trees, KNN, pruning, and imputation

Learning Day 7B - Classification Missing Coverage Bundle

Glossary Purpose

This glossary gives simple, beginner-friendly definitions for the main terms used in the Learning Day 7B missing classification coverage.

Term	Simple Meaning
Supervised learning	Learning from examples where the correct answer is available.
Classification	Predicting a category or label.
Regression	Predicting a continuous numeric value.
Feature	Input information used by the model.
Target	The correct answer the model learns to predict.
Class	One possible target category.
Binary classification	Classification with two possible labels.
Multiclass classification	Classification with more than two labels and one correct label per row.
Multilabel classification	Classification where one row can have multiple labels.
Train/test split	Separating data into learning data and evaluation data.
Stratify	Keeping class proportions similar in train and test sets.
Decision Tree	A model that predicts using if-then questions.
Root node	The first question in a Decision Tree.
Leaf node	The final prediction point in a Decision Tree.
Gini Index	A measure of how mixed a group is. Lower is cleaner.
Entropy	A measure of confusion inside a group. Lower is cleaner.
Information Gain	How much confusion a split removes. Higher is better.
Information Gain Bias	The problem where columns with many unique values look falsely useful.
Gain Ratio	A correction that reduces the unfair advantage of many tiny groups.
Hypothesis space	All possible models or trees the algorithm could choose.
Greedy search	Choosing the best option at the current step.
No backtracking	Not returning to change earlier tree decisions.
Inductive bias	The model preference that guides how it learns.
Occam Razor	Prefer the simpler explanation if performance is similar.
Decision boundary	The border between prediction areas.
Decision region	An area where the model gives the same prediction.
Overfitting	Model memorizes training data and performs worse on new data.
Pre-pruning	Stopping the tree early while it grows.
Post-pruning	Growing the tree first, then removing weak branches.
max_depth	Maximum allowed depth of the tree.
min_samples_split	Minimum rows needed before a node can split.

Term	Simple Meaning
min_samples_leaf	Minimum rows required in a final leaf.
ccp_alpha	Cost-complexity pruning strength in sklearn.
Subtree replacement	Replacing a branch with one final prediction.
Subtree raising	Moving a useful lower branch higher in the tree.
Error-based pruning	Removing branches that do not reduce error enough.
Complexity-based pruning	Removing branches that add too much complexity for little benefit.
MDL	Minimum Description Length; prefer shorter explanation if performance is similar.
KNN	K-Nearest Neighbors; predicts using similar old examples.
Lazy learner	A model that stores training examples and works mostly during prediction.
Eager learner	A model that learns rules or patterns during training.
k	Number of nearest neighbors used in KNN.
Neighbor	A similar old example used for KNN prediction.
Distance	How close or far two examples are.
Euclidean distance	Straight-line distance between numeric feature values.
Scaling	Putting numeric features on comparable scale.
StandardScaler	Centers and scales numeric features using mean and standard deviation.
One-hot encoding	Converts text categories into numeric 0/1 columns.
ColumnTransformer	Applies different preprocessing to different column groups.
Pipeline	Connects preprocessing and modelling steps in order.
SimpleImputer	Fills missing values using a simple rule such as median or most frequent.
KNNImputer	Fills missing numeric values using similar rows.
Majority voting	Every neighbor vote counts equally.
Distance-weighted voting	Closer neighbors get stronger votes.
Curse of dimensionality	Distance becomes less useful when there are too many features.
Precision	When the model predicts a class, how often it is correct.
Recall	Out of real class members, how many the model finds.
F1-score	Balanced score between precision and recall.
Confusion matrix	Table showing correct and wrong predictions by class.